

RESEARCH REPORT | REPORT 2

Waterflooding and Enhanced Oil Recovery: Displacement Efficiency and Method Screening

A technical review of mobility control, sweep, EOR screening criteria, and the limits of laboratory-to-field scaling

Scope: secondary and tertiary recovery in conventional reservoirs; screening guidance reflects published field experience and laboratory practice

Central conclusion

Recovery is the product of two independent efficiencies: how much oil is displaced where the injectant contacts the rock, and how much of the rock the injectant contacts at all. Most disappointing floods fail on the second term. A method that improves microscopic displacement while worsening sweep can reduce field recovery despite excellent core-flood results.

Executive summary

Primary depletion typically leaves the large majority of oil in place. Waterflooding, the dominant secondary method, restores reservoir energy and displaces oil toward producers, commonly lifting recovery into the range of 20 to 40% of original oil in place under favourable conditions. Enhanced oil recovery targets what remains: oil trapped by capillary forces at the pore scale, and oil in rock the flood never swept.

Overall recovery efficiency factorises into displacement efficiency and volumetric sweep efficiency, and volumetric sweep itself factorises into areal and vertical components. This factorisation is the central diagnostic tool of the discipline, because each term has different physical controls and responds to different interventions. Displacement efficiency is governed by capillary number, the dimensionless ratio of viscous to capillary forces, and improving it requires reducing interfacial tension by orders of magnitude. Sweep is governed by mobility ratio and geological heterogeneity, and improving it requires viscosifying the injectant, blocking thief zones, or re-engineering the well pattern.

Mobility ratio is the single most useful screening number in secondary recovery. Defined as the mobility of the displacing phase divided by that of the displaced phase, a value at or below unity produces a stable displacement front. Above unity, the displacing phase outruns the oil and fingers through it, and the excess is not recoverable by injecting more water: it produces water. Because viscosity ratio dominates the expression, heavy-oil waterfloods are frequently unfavourable from the first day of injection, which is the principal argument for thermal methods in those settings.

Screening principle

Diagnose which efficiency term is failing before selecting a method. A tracer or interference test that reveals channelling is diagnosing a sweep problem, and no reduction in interfacial tension will fix it. Conversely, residual oil saturation measured in swept rock is diagnosing a displacement problem, and no amount of conformance work will address it. Choosing an EOR method without this diagnosis is choosing at random.

Key findings

- Mobility ratio at or below 1.0 gives a stable front; above 1.0 the displacing phase fingers through the oil and produces water rather than incremental recovery.
- Mobilising waterflood residual oil requires increasing capillary number by roughly three to four orders of magnitude over typical waterflood conditions, which is achievable only through drastic interfacial tension reduction, not through higher injection rate.
- CO₂ miscible flooding achieves miscibility through multiple-contact mass transfer, not first-contact mixing; minimum miscibility pressure commonly falls in the range of roughly 1,200 to 4,500 psi depending on oil composition and reservoir temperature.
- Polymer flooding does not reduce residual oil saturation. It improves sweep by raising injectant viscosity, and its benefit is therefore bounded by how much unswept oil exists.
- Thermal methods dominate heavy oil because viscosity reduction with temperature is exponential and improves both mobility ratio and displacement simultaneously; SAGD has become the reference method for bitumen where vertical permeability supports steam-chamber growth.
- The Dykstra-Parsons coefficient quantifies permeability heterogeneity and predicts vertical sweep failure; values above roughly 0.7 indicate severe stratification in which conformance control precedes any EOR investment.

1. The efficiency factorisation

Overall recovery efficiency is the product of displacement efficiency, areal sweep efficiency, and vertical sweep efficiency. Each is a fraction between zero and one, and because they multiply, a strong result in one term cannot compensate for a weak result in another. A flood with 65% displacement efficiency in swept rock but 45% volumetric sweep recovers under 30% of the oil contacted at the pore scale by the flood's own laboratory metric.

Displacement efficiency measures the fraction of oil removed from the pore space that the injectant actually enters. Its ceiling is set by residual oil saturation, the immobile fraction trapped as disconnected ganglia by capillary forces at pore throats. Waterflood residual oil saturation commonly sits in the range of roughly 20 to 40% of pore volume in water-wet sandstone. That oil is not accessible by extending the flood; it is held by capillary forces that viscous forces at achievable field rates cannot overcome.

Areal sweep is controlled by well pattern, mobility ratio, and directional permeability. Vertical sweep is controlled by stratification, gravity segregation, and the density contrast between injectant and oil. Gas injection is particularly exposed on the vertical term: low-density gas overrides the oil column and sweeps the upper section preferentially, which is the physical basis for water-alternating-gas schemes that use water to suppress the override.

2. Mobility ratio and frontal stability

Mobility of a phase is its relative permeability divided by its viscosity. Mobility ratio compares the displacing phase behind the front to the displaced phase ahead of it. When mobility ratio is at or below unity, the displaced oil is at least as mobile as the water pushing it, and a perturbation on the front is self-correcting: the front remains sharp. When mobility ratio exceeds unity, a perturbation is self-amplifying, because water that runs ahead encounters higher water saturation and therefore higher water relative permeability, which accelerates it further. This positive feedback is viscous fingering.

The practical consequence is that unfavourable mobility ratio moves water breakthrough dramatically earlier and stretches recovery over a very large number of pore volumes injected, most of which is water recycling

through established fingers. The economic limit arrives at a water cut long before the displacement physics has exhausted the oil. This is why viscosity ratio, available from routine PVT data before any injection begins, is a first-order screening parameter rather than a detail.

Buckley-Leverett frontal advance theory provides the quantitative framework. It treats the flood as a one-dimensional hyperbolic displacement in which saturation propagates at a velocity set by the derivative of the fractional flow curve. The theory's characteristic multivalued saturation solution is resolved by the Welge tangent construction, which locates the shock front and gives average saturation behind it. The theory assumes incompressible immiscible flow, negligible capillary pressure, and homogeneous rock, so it is a reasoning tool and an upper bound rather than a field forecast.

3. Capillary number and the residual oil ceiling

The capillary number is the dimensionless ratio of viscous forces mobilising an oil ganglion to capillary forces trapping it, formed from Darcy velocity and viscosity divided by interfacial tension. Typical waterflood conditions produce capillary numbers around 10^{-7} to 10^{-6} , well below the threshold at which trapped oil begins to mobilise.

The capillary desaturation curve relates residual oil saturation to capillary number and is flat over the entire waterflood range, then falls steeply above a critical value commonly cited near 10^{-5} . That shape carries the central engineering implication of chemical EOR. Because injection velocity is bounded by fracture pressure and viscosity is bounded by injectivity, neither can be raised by the required orders of magnitude. Only interfacial tension can move that far, and that is precisely what surfactant flooding targets, aiming to reduce interfacial tension from roughly 20 to 30 dynes/cm to ultralow values on the order of 10^{-3} dynes/cm.

4. EOR method screening

EOR methods fall into three families distinguished by the physics they exploit. Thermal methods reduce oil viscosity with heat. Gas methods reduce or eliminate interfacial tension through mass transfer and miscibility. Chemical methods modify the injectant's viscosity, the interfacial tension, or both. Screening criteria below reflect published field and laboratory experience and function as a filter for candidate elimination rather than a substitute for evaluation.

Table 1. EOR method screening guidance

Method	Mechanism	Favourable conditions	Principal constraint
CO ₂ miscible	Multiple-contact miscibility; oil swelling; viscosity reduction	Light oil, roughly >27 API; depth sufficient to exceed MMP; MMP commonly ~1,200-4,500 psi	CO ₂ supply and cost; severe gravity override; corrosion; recycle handling
Hydrocarbon gas	Miscible or immiscible displacement; oil swelling	Light oil with available associated gas; good vertical containment	Opportunity cost of gas; override; requires reinjection infrastructure
Nitrogen / flue gas	Pressure maintenance; immiscible displacement	Deep, high-pressure, very light oil reservoirs	Very high MMP; typically immiscible in practice; low displacement efficiency
Polymer flood	Raises injectant viscosity; improves mobility ratio and sweep	Moderate viscosity oil; unfavourable mobility ratio; sufficient permeability	Does not reduce Sor; shear degradation; brine salinity and hardness limits; injectivity loss
Surfactant / ASP	Ultralow IFT mobilises residual oil; alkali generates in-situ soap	Light-to-medium oil; high residual oil in swept rock; controlled salinity	Chemical cost; adsorption losses on rock; emulsion breaking; scaling from alkali
Steam flood / CSS	Exponential viscosity reduction with temperature; distillation	Heavy oil, roughly <20 API; shallow; thick net pay to limit heat loss	Heat loss to overburden; steam-oil ratio economics; water treatment; emissions

Method	Mechanism	Favourable conditions	Principal constraint
SAGD	Gravity drainage of heated bitumen to a lower horizontal producer	Bitumen; thick, continuous, high vertical permeability; no top or bottom thief zones	Requires strong kv; shale barriers defeat chamber growth; high capital and energy intensity
In-situ combustion	Burns a fraction of oil in place to generate heat in the reservoir	Heavy oil where steam is uneconomic; sufficient oil for fuel deposition	Very difficult front control; unpredictable; limited sustained field success

Screening criteria summarise published field experience and laboratory practice. They are conditions under which methods have been observed to work, not thresholds that guarantee success; every candidate requires reservoir-specific laboratory work, pilot design, and economic evaluation.

5. Miscibility as a process, not a state

CO₂ and reservoir oil are not miscible on first contact at typical conditions. Miscibility develops through repeated contact as the injected gas moves through the reservoir, in a process that combines vaporising and condensing mass transfer: the gas extracts intermediate-molecular-weight components from the oil while depositing lighter components into it, and the two compositions converge until the interface vanishes. The minimum miscibility pressure is the pressure at which this convergence can complete.

Two consequences follow directly. First, MMP is a function of oil composition and reservoir temperature, not a property of CO₂, so it must be measured, conventionally by slim-tube test, and correlations serve only as screening estimates. Second, miscibility requires distance to develop, meaning near-wellbore regions may be effectively immiscible even in a flood classified as miscible. Reservoir pressure must be maintained above MMP throughout the swept region; falling below it anywhere converts that region to immiscible displacement and forfeits the recovery premium that justified the CO₂ cost.

CO₂ is roughly an order of magnitude less viscous than typical reservoir oil, so mobility ratio in a CO₂ flood is strongly unfavourable even when displacement efficiency approaches unity in the swept region. This is the central tension of gas EOR and the reason water-alternating-gas injection is the default field implementation: alternating slugs use water to reduce gas relative permeability and suppress both fingering and gravity override, trading some displacement efficiency for a large gain in sweep.

6. Heterogeneity and conformance

The Dykstra-Parsons coefficient quantifies permeability variation within a reservoir on a scale from zero, meaning perfectly homogeneous, to one, meaning infinitely heterogeneous. It is obtained by plotting permeability on a log-probability scale and comparing the median value with the value one standard deviation below it. Most real reservoirs fall roughly between 0.5 and 0.9, and values above roughly 0.7 indicate stratification severe enough that a high-permeability layer will dominate the flood.

This matters because heterogeneity and mobility ratio compound rather than merely coexist. A thief zone takes a disproportionate share of injection, and the resulting early breakthrough establishes a high-water-saturation channel with elevated water relative permeability, which takes an even larger share thereafter. Injecting an expensive chemical into that system routes it preferentially to the zone that least needs it. Conformance control, whether through gels, foams, or mechanical isolation, is therefore a prerequisite to EOR in stratified reservoirs, not an optional refinement.

7. Why laboratory results overstate field performance

A core flood measures displacement efficiency in a small, comparatively homogeneous sample under controlled conditions and near-perfect sweep. The field has heterogeneity spanning many orders of magnitude

in length scale, gravity acting over tens of metres of vertical section, chemical adsorption over kilometres of travel, and wells whose completions the laboratory never represented. Core-flood recovery is therefore an approximate upper bound on displacement efficiency and carries essentially no information about sweep.

Chemical EOR is the most exposed to this gap. Surfactant adsorption on rock surfaces is measured over centimetres in the laboratory but accumulates over the full inter-well distance in the field, and a slug that adsorbs to below its critical concentration stops working entirely from that point onward. Polymer shear degradation occurs in perforations and near-wellbore regions that no core flood reproduces. These are not calibration adjustments; they are mechanisms absent from the experiment, which is the argument for staged pilots between laboratory and full-field commitment.

8. Recommendations

- Compute mobility ratio from PVT and relative permeability data before any injection commitment, and treat a value materially above unity as a mobility-control problem requiring explicit design response.
- Diagnose whether the binding constraint is displacement or sweep using tracer tests, interference tests, and measured residual oil saturation in swept rock, and select the method family against the diagnosis.
- Measure MMP by slim-tube test for any CO₂ project; use correlations only for initial screening and state the pressure-maintenance requirement across the whole swept region as a design constraint.
- Quantify heterogeneity with the Dykstra-Parsons coefficient and address conformance before injecting chemicals into a stratified system.
- Design pilots to test the mechanisms core floods omit, specifically adsorption over field distances, shear degradation, gravity override, and areal conformance, rather than to re-confirm displacement efficiency.
- Model water-alternating-gas ratio and slug size explicitly against gravity number rather than adopting a default ratio from analogue fields with different vertical permeability.
- Track produced water composition and volumes as a leading indicator of channelling; a rising water cut at flat pressure is a sweep signal well before it becomes an economic one.

9. Limitations

Screening criteria and efficiency ranges reflect published field and laboratory experience and are not predictive for any specific reservoir. Buckley-Leverett theory is presented as a one-dimensional idealisation and does not represent gravity, capillarity, or heterogeneity. MMP correlations carry substantial scatter against measured slim-tube values, particularly for oils with unusual composition. EOR economics are omitted entirely and frequently dominate the technical screening in practice.

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